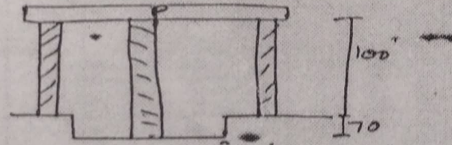


MID TERM I (B.E. III SEM 2025-26)
3ME44A(ME) Mechanics of Solids
MECH. ENGG. DEPT., MBM UNIVERSITY

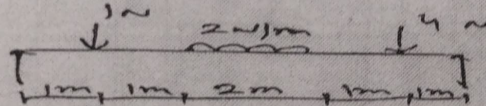
TIME: 60 MIN MM:10

2. Two brass rods and one steel rod together support a load as in fig.. If the stresses in bars and steel are not to exceed 60 N/mm^2 and 120 N/mm^2 . find the safe load that can be supported. $E_{\text{Steel}} = 2 \times 10^5 \text{ N/mm}^2$ $E_{\text{bars}} = 1 \times 10^5 \text{ N/mm}^2$, Area Steel = 1500 mm^2 , Area brass = 1000 mm^2



4. Draw the SFD and BMD for the following:

CO I (3)



5. Derive bending equation and state the assumptions made.

CO II (4)

OR

4. A beam of 6 meters span has to carry a load of 500 kg/m. Find the dimensions of beam, if the maximum permissible stress is limited to 80 Kg/cm^2 . The depth of beam is to be twice the width.

CO II (3)

Date: 23/09/2025

Exam Duration: 1 Hour

M.M.: 10

Solve the following questions. (Each Carry Equal Marks)

1) Solve:

$$\frac{dx}{dt} = ny - mz, \quad \frac{dy}{dt} = lz - nx, \quad \frac{dz}{dt} = mx - ly$$

Or

$$\frac{dx}{x} = \frac{dy}{y} - \frac{dz}{z - a\sqrt{x^2 + y^2 + z^2}}$$

2) Solve:

$$\frac{dx}{dt} + 7x - y = 0$$

$$\frac{dy}{dt} + 2x + 5y = 0$$

Or

$$t \frac{dx}{dt} = t - 2x$$

$$t \frac{dy}{dt} = tx + ty + 2x - t$$

3) Solve:

$$3x^2 dx + 3y^2 dy - (x^3 + y^3 + e^{2z}) dz = 0 \text{ (By any Method)}$$

Or

$$(y^2 + yz) dx + (xz + z^2) dy + (y^2 - xy) dz = 0 \text{ (By any Method)}$$

4) Solve:

$$(p^2 + q^2) y = qz \text{ (By Charpits Method)}$$

Or

$$r - t \cos^2 x + p \tan x = 0 \text{ (By Monge's Method)}$$

1st Midterm

ME 201A: Engineering Thermodynamics _Date 23/09/2025 Exam Duration: One Hour (2:30 to 3:30 PM)

M.M.:20

CO1	1. Write zeroth law of thermodynamics. ✓	[1]
CO1	2. Write two corollaries of second law of thermodynamics. -	[2]
CO1	3. Derive the expression of energy balance for unsteady flow process (filling a tank). [2] -	
CO2	4. State inequality of Clausius. [2] ✓ -	
CO2	5. Prove that maximum work obtainable from a finite body and a TER is - $W_{\max} = C_p[(T - T_0) - T_0 \ln(\frac{T}{T_0})]$ where symbols have usual meaning. ✓✓	[5]
CO2	6. State the entropy principle. [1]	
CO4	7. What is stoichiometric air and excess air. Write the combustion equation for fuel C_8H_{18} . Find the air fuel ratio of fuel C_8H_{18} . [2+2+3]	

MBM UNIVERSITY, JODHPUR

1st Mid Term

II Year III Semester 2025-26

3ME45A: Production Machine Tools

Date: 25/09/2025

Exam Duration: 1 Hour

M.M.: 10

- ✓ 13. a) With the help of neat sketch explain the working principle of lathe machine tool. ✓
b) Differentiate between up milling and down milling. ✓
- ✓ 14. a) Differentiate between Capstan & Turret lathe with Engine Lathe. ✓
b) Write the applications of shaper machines in a typical workshop also compare shaper and planner in terms of the operations, and type of Work pieces. ✓
- ✓ 15. a) With the help of neat labelled diagram explain the working principle of Column and Knee type shaper machine tool.

OR

- b) With the help of neat labelled diagram explain Quick Return Mechanism. ✓

MBM UNIVERSITY, JODHPUR
1st Mid Term
II Year III Semester 2025-26
Subject: 3ME42A(ME): Material Technology

Date: 24/09/2025

Time: 1 Hr.

M.M.: 10

All questions compulsory:

- | | | | |
|----------------|-----|--|---|
| Q.1 | CO1 | Describe different crystal structure with diagram. | 3 |
| Q.2 | CO1 | Copper has F.C.C. structure and the atomic radius $1.278\text{\AA}$. Calculate the density of copper crystal.
Given atomic weight of copper = 63.5. Avogadro no. = 6.023×10^{23} . (Ans x) | 4 |
| Q.3 | CO2 | Describe the role of <u>dislocation</u> in metals and alloys with diagram. | 3 |

MBM University, Jodhpur
Department of Mechanical Engineering
B.E. IInd Year IIIrd Semester
First Mid-Term Examination Session 2025-26
3ME43A(ME): Kinematics of Machines

Time: 1 hour

Maximum Marks – 10

5. (a) Name the materials of flat belts, V-belts and ropes. ✓ (1)
- (b) What is meant by cross-belt drive? Find the length of belt in a cross-belt drive. ✓ (2)
- (c) A casting having a mass of 100 kg is suspended freely from a rope. The rope makes 2 turns a drum of 300 mm diameter rotating at 24 rpm. The other end of the rope is pulled by a man. Calculate the force required by the man, power to raise the casting and the power supplied by drum run by a prime-mover. Take $\mu = 0.3$. (3)
6. (a) Discuss various types of constrained motion. ✓ (1)
- (b) Sketch and explain the various inversion of single -slider crank chain mechanism. ✓ (2)
- (c) Define Grashof's law. State how is it helpful in classifying the four-link mechanism into different types. (1) ✓